

Computer Science Fundamentals

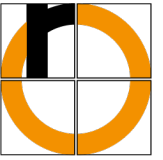
Cryptography – Classical Methods

Technische Hochschule Rosenheim
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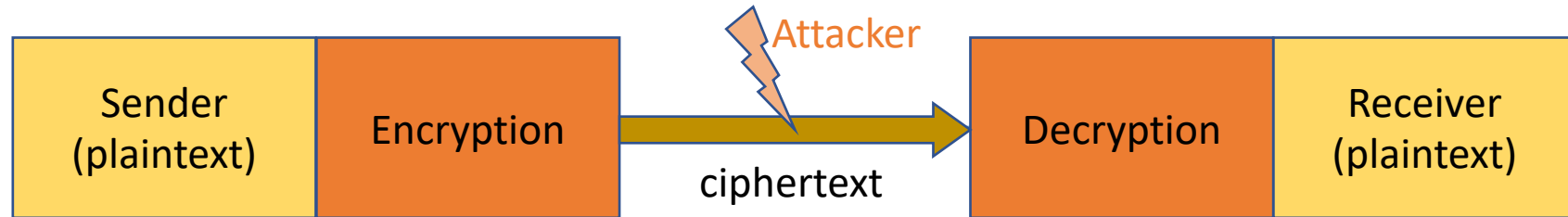
- What is a cryptosystem?
- Some classical methods of cryptography

Encrypted transmission of messages is of great interest

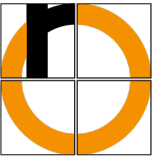
- not only for the military and secret services
- but also for companies (e.g., transmission of confidential information on new products)
- and individuals (e.g., Online-Banking → https)



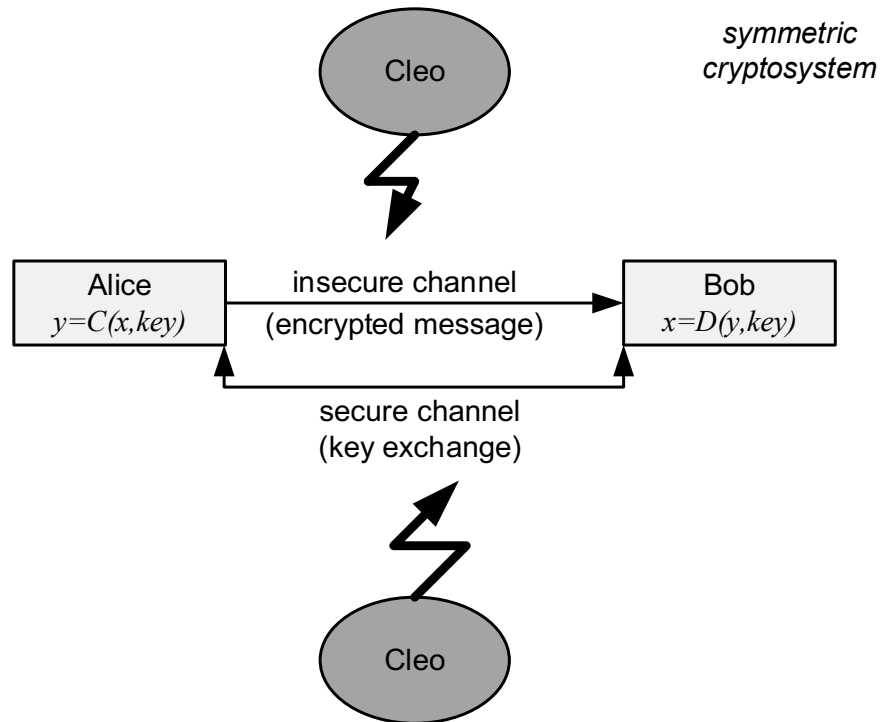
Process



- Encryption of the message (called **plaintext**) into a **ciphertext**
 - Use of encryption algorithm
 - and key parameters
- Sender sends ciphertext to the recipient
- Decryption of the ciphertext by recipient
 - Use of a suitable decryption algorithm
 - and the same key parameters
- Recipient gets the plaintext of the message

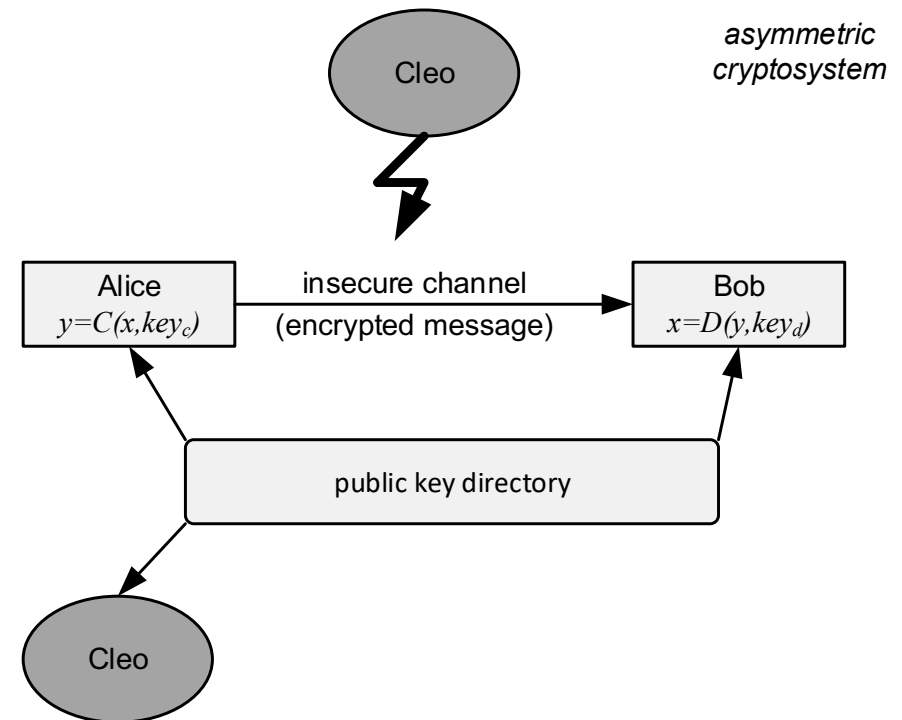


Symmetric Encryption



- same secret key for encryption & decryption
- key exchange via secure channel

Asymmetric Encryption



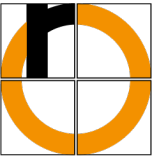
- encryption using public key of recipient
- decryption using private (secret) key of recipient

- Formulated 1883
- Principle of all modern cryptographic methods
- Security of a method
 - must **not** be based on secrecy of the algorithm
 - but on secrecy of the key
- This means
 - No „Security through Obscurity“
 - Algorithms are public

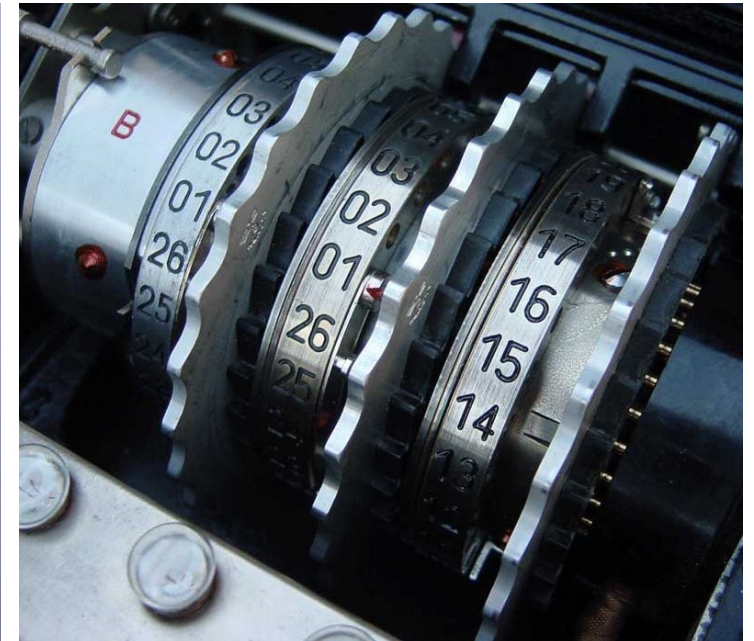
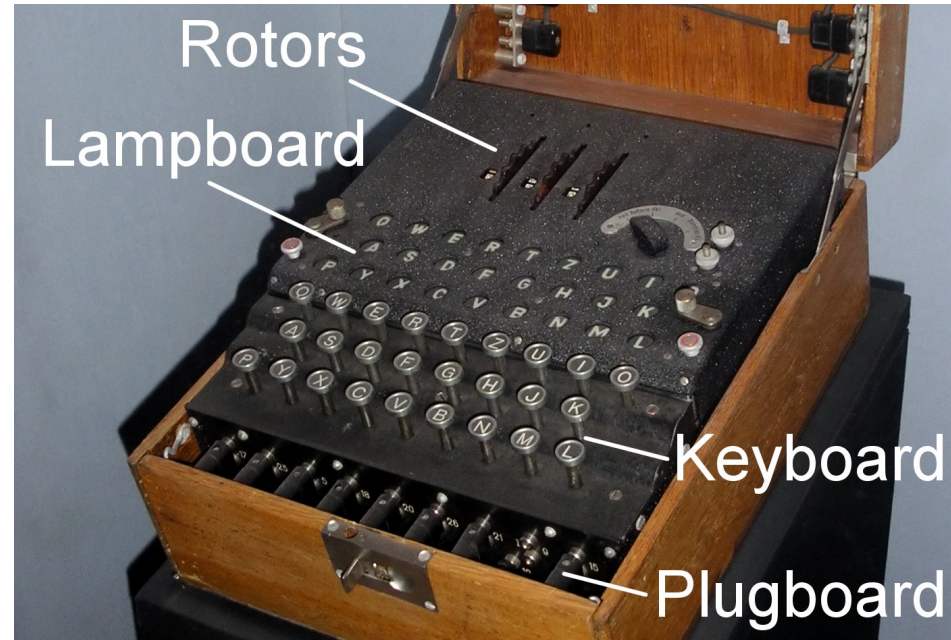
- classical = developed before 1950
- presented here to illustrate the basic encryption principles: **substitution ciphers**
- classical methods in pure form are no longer in use today
 - but they are a part of modern ciphers like AES

- Idea: Replace (“substitute”) units of plaintext by corresponding units of ciphertext
- Types of substitution ciphers:
 - **simple** substitution: replace each letter of the plaintext alphabet by (always the same) corresponding letter of the ciphertext alphabet (**bijective** mapping).
 - **homophonic** substitution: replace each letter of the plaintext alphabet by one of the corresponding symbols of the ciphertext alphabet; **one** plaintext letter can be mapped **to many** different ciphertext symbols (**homophones**).
 - **polyalphabetic** substitution: like “simple”, but **different mappings** are used for each position in the plaintext using a defined algorithm, e.g., periodically.
 - **polygram** substitution: instead of single letters, **replace whole blocks** of letters.
- Breakthrough of these methods with the availability of electromechanical encryption machines (like the **Enigma**, which uses polyalphabetic substitution)

Enigma



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- Example: Use an affine transformation

- Symbols x_i of an alphabet A containing a total of n symbols are mapped to the same alphabet using

$$x_i \longrightarrow x_{(k \cdot i + d) \bmod n}$$

- **k** is the **multiplicative key**
 - **d** is the **additive key**
-
- Special cases
 - $k = 1 \longrightarrow$ Caesar Code
 - $d = 0 \longrightarrow$ Product Ciphers

- Alphabet A with n symbols
- symbol in ciphertext =
 - Multiply the position i of a symbol in the alphabet by key k,
 - add d,
 - and reduce mod n.
- Arbitrary combination of key parameters not possible for unambiguous mapping
- Example: $k = 4, n = 26, d = 0$

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	E	I	M	Q	U	Y	C	G	K	O	S	W	A	E	I	M	Q	U	Y	C	G	K	O	S	W

- Repetitions occur – unsuitable!

- For a suitable combination (k, n) :
 - k and n must be **relatively prime**: $\gcd(k, n) = 1$
 - Only these keys k are suitable, as they have a **modular inverse** k^{-1} with $k \cdot k^{-1} \bmod n = 1$
 - For computing the modular inverse use
 - extended Euclidean algorithm
 - Euler's/Fermat's theorem
 - details see appendix/math's course
- Example: relatively prime to $n = 26$ are $\{1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25\}$
 - Therefore: $k = 7$ is suitable
 - Inverse mod 26: $7^{-1} \equiv 15$
 - Test: $7 \cdot 15 \bmod 26 = 105 \bmod 26 = 1$

Simple (Affine) Substitution Ciphers – Example

Encryption of plaintext *COMPUTER*

- with multiplicative key $k=7$

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	H	O	V	C	J	Q	X	E	L	S	Z	G	N	U	B	I	P	W	D	K	R	Y	F	M	T

- and additive key $d=5$

Encryption

- plaintext : C O M P U T E R
- multiplication ($k=7$): O U G B K D C P
- shift ($d=5$): T Z L G P I H U

Decryption with inverse operations

- ciphertext: T Z L G P I H U
- shift ($-d = -5$): O U G B K D C P
- multiplication ($k^{-1}=15$): C O M P U T E R

- This is a **polyalphabetic substitution** cipher
- Generalization of Caesar cipher: use multiple substitution tables (with different shifts)
- Define a **key** that is **built from letters** of the plaintext alphabet
 - The key defines the parameter **d** that is used as offset for shifting each letter
- To obtain the letter index of the ciphertext
 1. match positions of repeated key and plaintext
 2. for each position:
 - the index of the plaintext letter is added to
 - the index of the letter in the key

Example:

Latin alphabet (0-25), mod 26, key = **BCD**

- plaintext : **SECRETTEXT**
- key (shift): **BCDBCDBCDB**
- ciphertext: **TGFSGWUGAU**

- Breaking the Vigenère cipher is getting harder the longer the key
- If the key
 - has the same length as the plaintext,
 - is completely random,
 - and never reused.

we get the **Vernam cipher** or **one-time pad**

- For one-time pads we typically use the alphabet $\{0, 1\}$
 - the plaintext is converted to binary,
 - the key is a random sequence of bits,
 - the mod 2 addition becomes a simple bitwise XOR of plaintext/ciphertext and key for en-/decryption
- Example:

Encryption

- plaintext: 1 0 0 1 1 0 0 1
- key: 0 1 1 1 0 0 1 1 **XOR**
- ciphertext: 1 1 1 0 1 0 1 0

Decryption

- ciphertext: 1 1 1 0 1 0 1 0
- key: 0 1 1 1 0 0 1 1 **XOR**
- plaintext: 1 0 0 1 1 0 0 1

- One-time pads offer **perfect secrecy**
 - The encrypted data do **not** allow **any** conclusions to be drawn about the plain text except for its length
 - **One-time pads cannot be broken** – no matter how much computing power is invested
 - Proof by Shannon 1949

Claude E. Shannon: Communication Theory of Secrecy Systems. In: *Bell System Technical Journal* 28(4). AT&T Corporation, USA, October 1949, pp. 656–715
<http://pages.cs.wisc.edu/~rist/642-spring-2014/shannon-secrecy.pdf>
- Practical limitations
 - this only applies if the key is generated from real random numbers
 - pseudo-randomness is not sufficient
 - the key is as long as the data and has to be exchanged via a secure channel
→ used very rarely

Example

- plaintext x : **1 0 0 1 1 0 0 1**
- key k : **0 1 1 1 0 0 1 1** **XOR** if the key is random...
- ciphertext y : **1 1 1 0 1 0 1 0** ... the ciphertext appears to be random as well

For each bit y_i of the ciphertext we obtain an equation:

$$y_0 \equiv x_0 + k_0 \pmod{2}$$

$$y_1 \equiv x_1 + k_1 \pmod{2}$$

$$y_2 \equiv x_2 + k_2 \pmod{2}$$

...

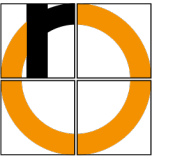
An attacker would have to solve these equations for x_i

Attacker knows the value for y_0 .

If k_0 is chosen from a truly random uniform distribution:

- The solutions 0 and 1 for x_0 are equally likely (50%)
- There is no dependency between k_0 and any other bit of the key,
- and therefore no dependency to any other equation

→ the system of equations can not be solved



Appendix: Modular Inverse

- $\gcd(a, b) = \gcd(b, a \bmod b)$ *(in German: ggT)*
 - for $a \geq b$
 - Stop if $b = 0$
 - then a is the gcd
- Examples:
 - $\gcd(26, 13) = \gcd(13, 0) \rightarrow \gcd = 13$
 - $\gcd(26, 7)$
 - $= \gcd(7, 5)$
 - $= \gcd(5, 2)$
 - $= \gcd(2, 1)$
 - $= \gcd(1, 0) \rightarrow \gcd = 1$

For determining the modular inverse

- $\gcd(a, b) = s \cdot a + t \cdot b$
 - s, t are integers
 - if $\gcd(a, b) = 1 \Rightarrow t$ is the (multiplicative) modular inverse of $b \pmod{a}$
- Example: modular inverse of 7 mod 26

$$\begin{array}{llll} 26 & = 3 \cdot 7 + 5 & \longrightarrow 5 & = 26 - 3 \cdot 7 \\ 7 & = 1 \cdot 5 + 2 & \longrightarrow 2 & = 7 - 1 \cdot 5 = 7 - (26 - 3 \cdot 7) = \\ & & & = -26 + 4 \cdot 7 \\ 5 & = 2 \cdot 2 + 1 & \longrightarrow 1 & = 5 - 2 \cdot 2 \\ & & & = 26 - 3 \cdot 7 - 2 \cdot (-26 + 4 \cdot 7) \\ & & & = 3 \cdot 26 - 11 \cdot 7 \end{array}$$

Inverse exists

and equals $-11 \equiv 15 \pmod{26}$

- The function's value is the number of natural numbers
 - that are smaller than n
 - and are relatively prime to n
 - $\phi(n) = |\{1 \leq x \leq n \mid \gcd(x, n) = 1\}|$
- Computation (p, q are prime numbers $p \neq q$)
 - $\phi(p) = p - 1$ all integers from 1 to $p - 1$ are relatively prime to p
 - $\phi(pq) = \phi(p)\phi(q) = (p - 1)(q - 1)$
 - $\phi(p^i) = p^{i-1}(p - 1)$
 - $\phi(p^i q^j) = \phi(p^i)\phi(q^j) = p^{i-1}(p - 1) q^{j-1}(q - 1)$
- Examples
 - $\phi(5) = 4$
 - there are four numbers < 5 that are relatively prime to 5, namely 1, 2, 3, 4
 - $\phi(15) = \phi(3 \cdot 5) = \phi(3)\phi(5) = 2 \cdot 4 = 8$
 - $\phi(27) = \phi(3^3) = 3^2 \cdot (3 - 1) = 9 \cdot 2 = 18$
 - the numbers that are relatively prime to 27 are: 1, 2, 4, 5, 7, 8, 10, 11, 13, 14, 16, 17, 19, 20, 22, 23, 25, 26
 - $\phi(72) = \phi(2^3 \cdot 3^2) = 2^2 \cdot (2 - 1) \cdot 3^1 \cdot (3 - 1) = 4 \cdot 3 \cdot 1 \cdot 2 = 24$

- **Euler's theorem:**
for all $x \in \mathbb{Z}, n \in \mathbb{N}, \text{ggT}(x, n) = 1$:

$$x^{\phi(n)} \bmod n = 1$$

- Special case: n is a prime number $p \rightarrow$ **Fermat's** little theorem:

$$x^{p-1} \bmod p = 1$$

- It holds:

$$x \cdot x^{\phi(n)-1} \bmod n = 1$$

and therefore:

$$x^{-1} = x^{\phi(n)-1} \bmod n$$

or, for a prime:

$$x^{-1} = x^{p-2} \bmod p$$

- Using a prime number as module: $p = 31$
 - wanted: modular inverse for $x = 2$
 - it holds: $2^{-1} \equiv 2^{31-2} \bmod 31 = 2^{29} \bmod 31 = 16$
 - Test: $2 \cdot 16 = 32 \bmod 31 = 1$
- With $n = 26$
 - wanted: modular inverse for $x = 7$
 - determine $\phi(26)$, i.e., the **number** of positive integers that are relatively prime to 26.
 - Prime factorization:
$$26 = 13 \cdot 2$$
 - therefore: $\phi(26) = \phi(13)\phi(2) = 12 \cdot 1 = 12$
 - it holds: $7^{-1} \equiv 7^{12-1} \bmod 26 = 7^{11} \bmod 26 = 15$
 - Test: $7 \cdot 15 = 105 \bmod 26 = 1$